Available BSc Projects

BIOPHOTONICS AND MEDICAL IMAGING GROUP
Department of Physics and Astronomy,
VU University, LASERLAB

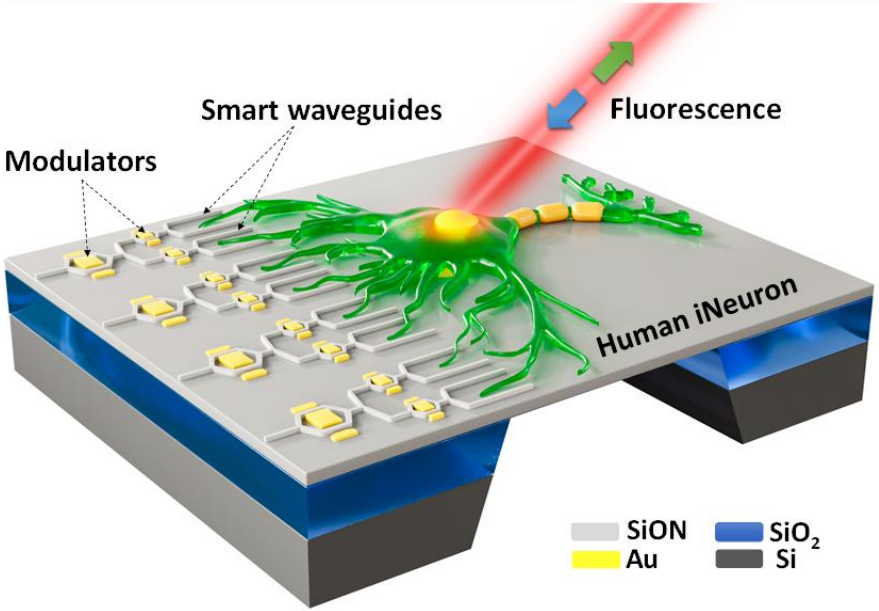
Stimulating biological neurons with photonic microchips (Computational project)

Neural signal processing in humans differs fundamentally from that in traditional animal models, yet many of the underlying mechanisms remain poorly understood. In particular, how the physical properties of dendrites and synapses shape the input–output behavior of human neurons is still an open question. In this project, we will develop a novel experimental approach to stimulate individual dendrites using a custom-designed photonic microchip. This work sits at the intersection of physics, photonics, and neuroscience, and involves hands-on experience with microchip fabrication, optical stimulation, and signal analysis. The project is carried out in collaboration with NeuroSpector (a spin-off from the VU Neuroscience Department), with close guidance from an experienced postdoctoral researcher who will support the simulations and design of the microchips.

Student Tasks:

- **Design** electrodes and waveguides to be used in the photonics microchip.
- **Perform optical simulations** to design electrodes and waveguides.

This project offers a **unique opportunity** to work at the intersection of physics, engineering, and biology, contributing to cutting-edge research in neuroscience.



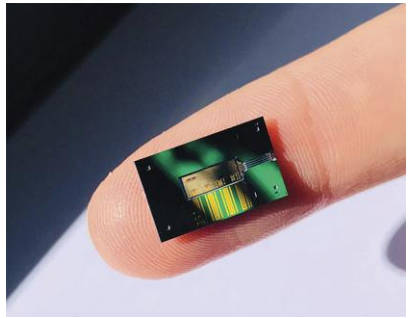


Fig. 1. Schematic of the photonic microchip for neuron stimulation (top) and the fabricated photonic chip (bottom).

Contact person: Assoc. Prof. B. Imran Avci

Email: b.i.avci@vu.nl

Stimulating biological neurons with photonic microchips (Characterization project)

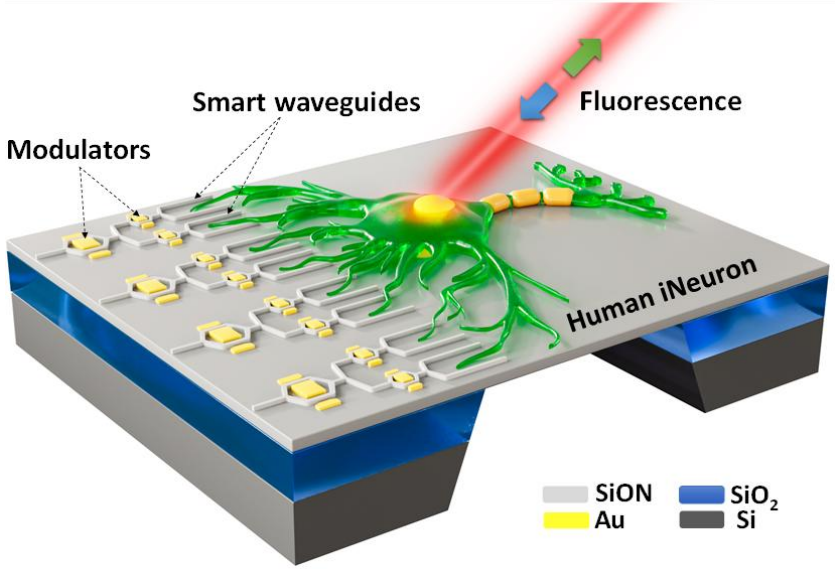
Neural signal processing in humans differs fundamentally from that in traditional animal models, yet many of the underlying mechanisms remain poorly understood. In particular, how the physical properties of dendrites and synapses shape the input–output behavior of human neurons is still an open question. In this project, we will develop a novel experimental approach to stimulate individual dendrites using a custom-designed photonic microchip. This work sits at the intersection of physics, photonics, and neuroscience, and involves hands-on experience with microchip fabrication, optical stimulation, and signal analysis. The project is carried out in collaboration with NeuroSpector (a spin-off from the VU Neuroscience Department), with close guidance from an experienced postdoctoral researcher who will support both the fabrication of the chips and the experimental work.

Student Tasks:

- **Perform optical measurements** on fabricated photonic microchips.
- **Analyze the results** to understand the behaviour of the fabricated devices.

This project offers a **unique opportunity** to work at the intersection of physics, engineering, and biology, contributing to cutting-edge research in neuroscience.

Contact person: Assoc. Prof. B. Imran Avci
Email: b.i.avci@vu.nl



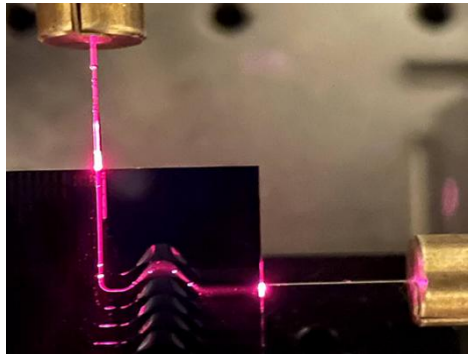


Fig. 1. Schematic of the photonic microchip for neuron stimulation (top) and the characterization of the devices (bottom).

Stimulating biological neurons with photonic microchips (Fabrication)

In a new neuroscience project, we are developing microfluidic platforms to study how neurons process signals in controlled, human-relevant environments. The microfluidic devices allow neurons to be cultured and guided within well-defined geometries, enabling precise investigation of neuronal connectivity and signal propagation. Using rapid, in-house fabrication, we can design and test new microfluidic layouts within hours, allowing fast iteration and hands-on exploration of neural signal processing.

Student Tasks:

- Design the microfluidics devices
- Fabricate microfluidics devices usign our custom direct laser writer
- Experimentally characterize the fabricated devices.

This is an exciting opportunity for students interested in **nanofabrication, optics, and integrated photonics** to contribute to a high-impact research project.

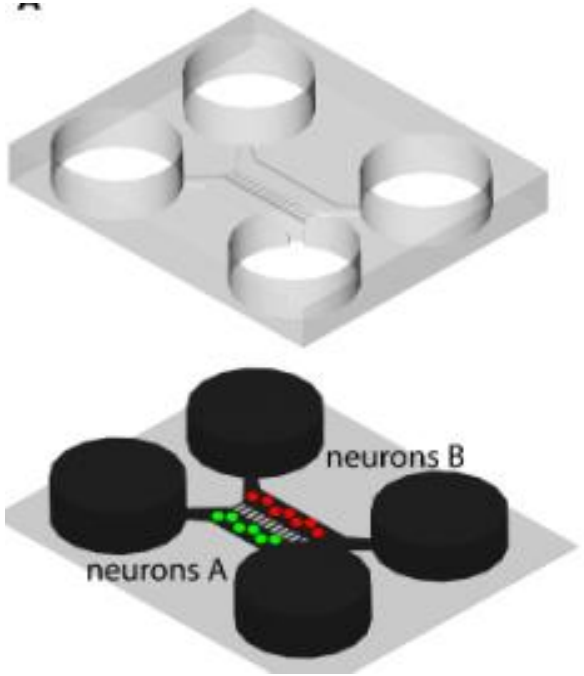


Fig. 1. Microfluidic devices to be tailored for growing neurons and separating their dendritic branches from axons.

Contact person: Assoc. Prof. B. Imran Avci
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Evaluate illumination through a diffuser for holographic OCT

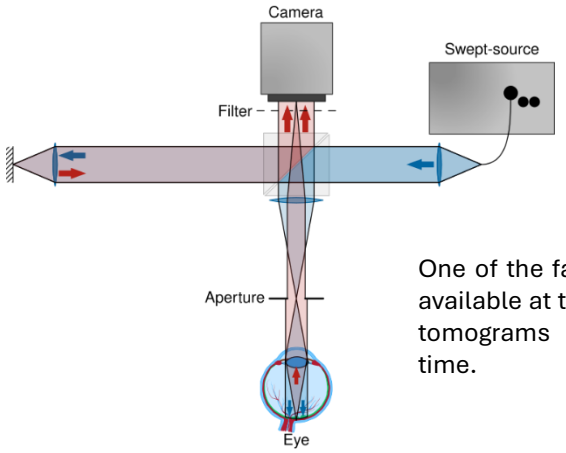
Full-field Fourier-domain optical coherence tomography (FF-FD-OCT) is an emerging three-dimensional, tomographic technique that achieves record-breaking imaging speed at about **100 MHz A-scan (depth scan) rate**. It obtains this by parallelizing the acquisition in lateral and axial direction, e.g., by using one of the fastest available high-speed area cameras in the world. One of its applications is retinal imaging, where speckle noise disturbs imaging quality. Within this project, we want to evaluate

- If speckle noise can be mitigated by placing a diffuser in the illumination arm.
- How this scenario behaves for single mode and multimode illumination.
- What kind of diffuser is optimally suited for speckle suppression.

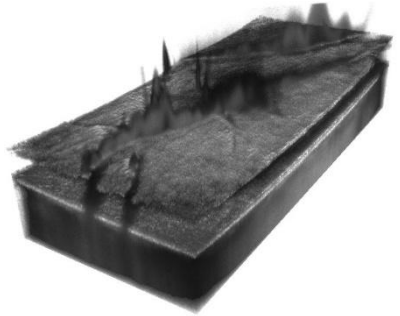
This project requires data acquisition and data processing with **Python**.

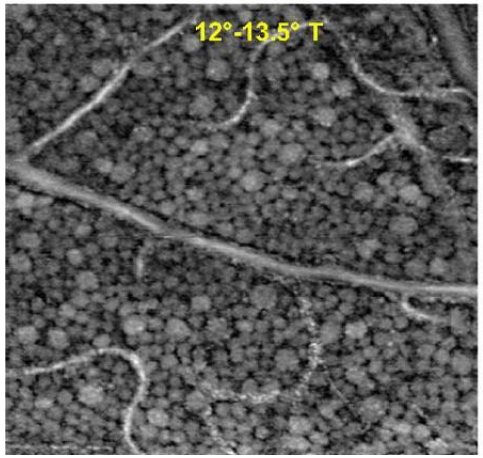
Our ultimate goal for this technique is to promote FF-FD-OCT as the **next-generation retinal imaging technique** providing **functional** and **morphological** contrast and imaging neurons in the living retina.

For questions, more details, or further references, please contact Dr. Dierck Hillmann (d.w.a.hillmann@vu.nl)



One of the fastest OCT systems s available at the VU can acquire 3D tomograms of tissue in record time.





Ganglion cells (retinal neurons), imaged with special data acquisition and processing using optical coherence tomography. Obtaining comparable images requires suppression of speckle noise. Can we obtain similar results from fewer images?

Image source: Liu et al., PNAS 114(48), 2017

Contact: Dr. Dierck Hillmann
d.w.a.hillmann@vu.nl

A Simulation of FF-FD-OCT using Scattering Theory

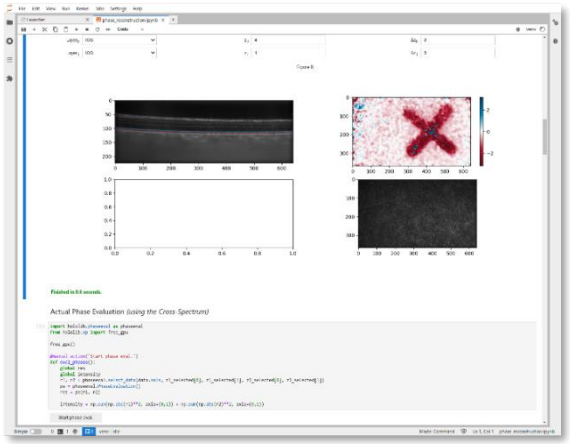
Full-field Fourier-domain optical coherence tomography (FF-FD-OCT) is an emerging three-dimensional, tomographic technique that achieves record-breaking imaging speed at about **100 MHz A-scan (depth scan) rate**. It obtains this by parallelizing the acquisition in lateral and axial direction, e.g., by using one of the fastest available high-speed area cameras in the world. A proper OCT data simulation requires knowledge of the imaging scenario and the physics behind the approach. We are looking to simulate this scenario.

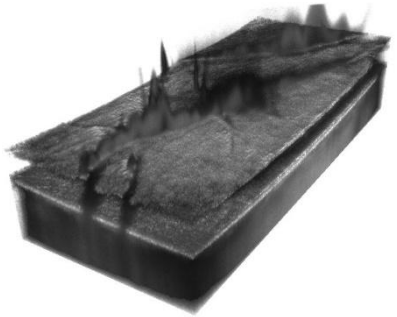
- Creating a forward simulation of an OCT system and expanding it to include more and more free parameters.
- Allow easy access to data that can be used to develop and test reconstruction algorithms without requiring measurements.
- Take defocus, speckle, and other image phenomena of OCT into account.

The projects requires understanding the physics behind the signals and performing numerical tasks in **Python**. Ideally, the project should provide a simulation framework.

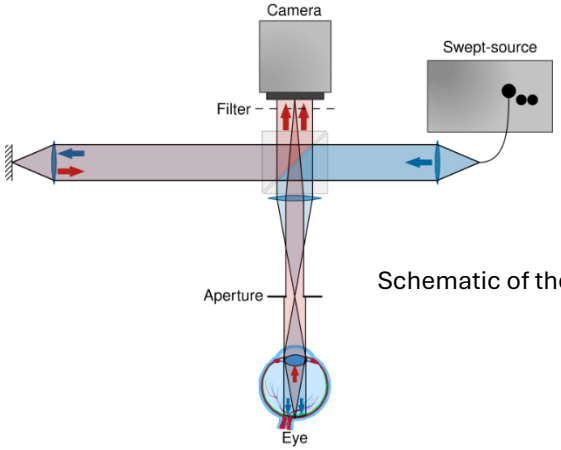
Our ultimate goal for this technique is to promote FF-FD-OCT as the **next-generation retinal imaging technique** providing **functional** and **morphological** contrast and imaging neurons in the living retina.

For questions, more details, or further references, please contact Dr. Dierck Hillmann (d.w.a.hillmann@vu.nl)





A solid infrastructure for handling full-field OCT data (e.g., reconstruction and phase evaluation) on high GPU hardware are available and allow evaluation of state-of-the art techniques in this project.



Schematic of the system being simulated.

Contact: Dr. Dierck Hillmann
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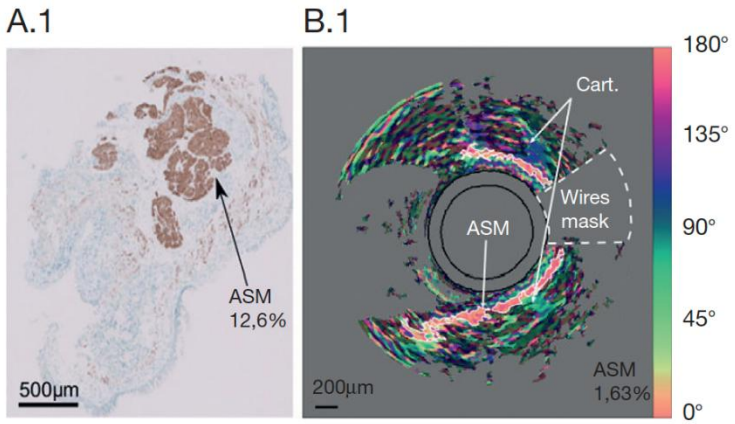
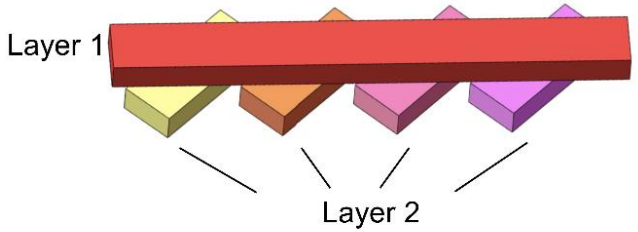
Development and characterization of a birefringent phantom for PS-OCT imaging

- Description: Polarization Sensitive Optical Coherence Tomography (PS-OCT) uses the polarization properties of light to measure the presence and orientation of fiber structures in the lungs of patients with lung diseases such as asthma and ILD. The methods for determining the polarization properties of a sample require complex mathematical derivation and thus can benefit from validation through a well characterized phantom. We want to develop a phantom with known polarization properties, to test and validate the PS-OCT processing algorithm.

- Goal: In this research project, you will work on making a birefringent phantom and characterizing it by scanning the phantom with our PS-OCT imaging system and processing with our new PS-OCT processing algorithm.



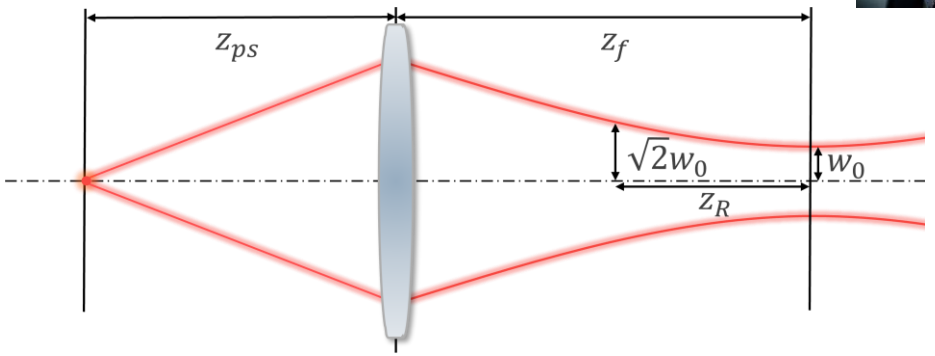


1. Vaselli M, Wijsman PC, Willemse J, Goorsenberg AWM, Feroldi F, d'Hooghe JNS, Annema JT, de Boer JF, Bonta PI. Polarization Sensitive Optical Coherence Tomography for Bronchoscopic Airway Smooth Muscle Detection in Bronchial Thermoplasty-Treated Patients With Asthma. Chest. 2021 Aug;160(2):432-435.
2. X. Liu, K. Beaudette, X. Wang, L. Liu, B. Bouma, and M. Villiger, "Tissue-like phantoms for quantitative birefringence imaging," Biomed. Opt. Express 8, 4454-4465 (2017).

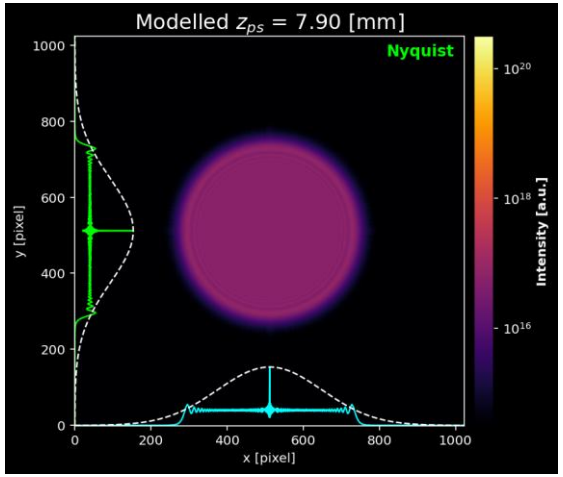
Contact: Prof. Johannes F. de Boer (j.f.de.boer@vu.nl)
Jules Blom (j.blom@vu.nl)

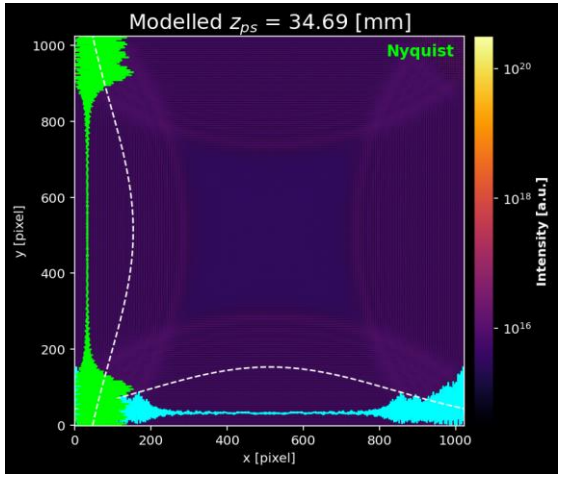
Development and characterization of a beam profile from an imaged point source at various depths

- **Description:** Targeted Depth Resolved Fluorescence (TDRF) is a technique that uses tailor-made fluorescent particles that bind to, for example, carcinoma cells to aid in detecting onset esophageal cancer within patients. To be able to do this, the beam profile of a point source at various depths needs to be characterized and fully understood. A theoretical model has already been developed and a section of the experimental setup has been built. You will help with the finishing touches to this setup, enabling you to perform measurements and compare these to the theoretical data.



- **Goal:** In this research project, you will work on further developing an experimental setup which is a 10x scale of a to be build targeted depth resolved fluorescence endoscopic imaging system which is to be used during *in vivo* esophageal carcinoma research.





- **You:** Enjoy **optics** and understand the underlying physics, are comfortable with **Python** and like a well-balanced research project combining experiment in the lab with theory behind a desk.

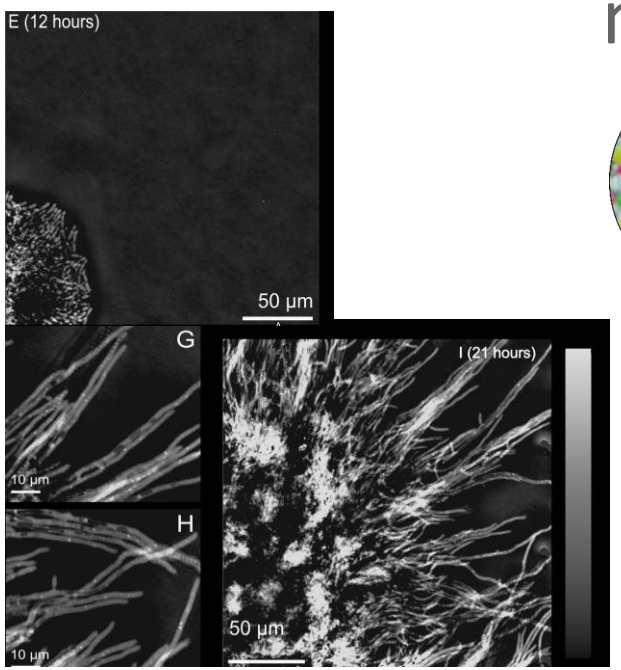
Contact: Prof. Johannes F. de Boer (j.f.de.boer@vu.nl)

Louis Veerkamp (a.t.g.veerkamp@vu.nl)

matzlab

Physics informed image segmentation

Modern image segmentation, the task of turning images of complex biological structures into distinct components, is currently performed using AI-models.
To train these models, large libraries of annotated data are necessary which are difficult and laboursome to obtain, we want to eliminate this bottleneck.



Project:

We want to explore if and how we can train segmentation algorithms based on in-silico generated images. Towards this goal, we are looking for students that would be excited to dive into the morphologies of relevant microbes, build simple computational models to describe these structures and then turn this information into computer generated images to eliminate the need for manual annotation. Your work would be closely integrated into a collaborative project with the Department of Microbiology (Evelina Tutucci) where we are working on one of the most relevant pathogenic fungi: C. Albicans.

Contact: Matz Liebel
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Development of a correction method for Polarization Mode Dispersion in PS-OCT

Description: Polarization Sensitive Optical Coherence Tomography (PS-OCT) is an extension of OCT combining structural images with tissue specific contrast by assessing tissue birefringence and optic axis (OA) orientation. In lung tissue e.g. light experiences different refractive indices depending on its polarization state; an effect called *birefringence*. PS-OCT systems are often built with single-mode optical fibers (SMFs) imper-fections in the fiber core that change the polarization state of the OCT light to an unknown state in a *k*-dependent manner. This effect is known as Polarization Mode Dispersion (PMD); it adds significant amounts of noise and causes erroneous local birefringent changes in PS-OCT measurements.

Goal: In this project you will further develop and validate a novel method for correcting PMD. You will use an extended Jones-Mueller matrix analysis to investigate the eigensystem of the PS-OCT signal and apply a new Jones optimization method to compensate for *k*-dependent PMD. The method will be tested on tissue simulations and real measurements.

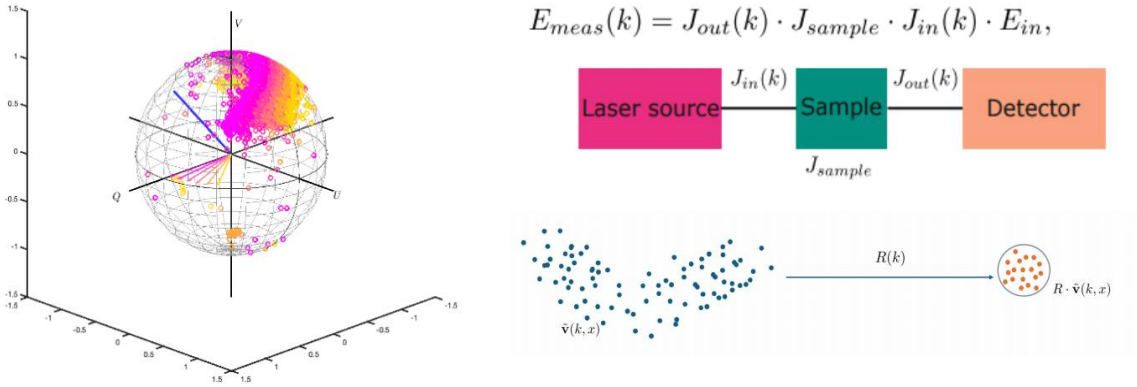


Figure 1: The left panel shows the optic axes of a simulated a sample and the effect of the k-dependent PMD as a trace of the eigenvectors around the sample optic axis (shown in blue). The top right panel is the simplified mathematical matrix model for the simulated sample. To compensate for the k-dependent PMD effects we use a novel eigenvector optimization method.

Contact: Prof. Johannes F. de Boer (j.f.de.boer@vu.nl)
dr. Patrick González (p.j.gonzalez@vu.nl)

AI enhanced PS-OCT

- **Description**: Polarization sensitive OCT (PS-OCT) is an extension of OCT that allows for imaging of tissue birefringence. A key part of the image reconstruction algorithm, is a novel method we have developed for the compensation of polarization mode dispersion (PMD) inherent to PS-OCT systems. This method relies on an extended Jones- and Stokes eigenvector analysis to reconstruct the tissue eigensystem. However, speckle noise can lead to an unreliable reconstruction of the tissue eigensystem.

Optic Axis with PMD

Input Optic Axis

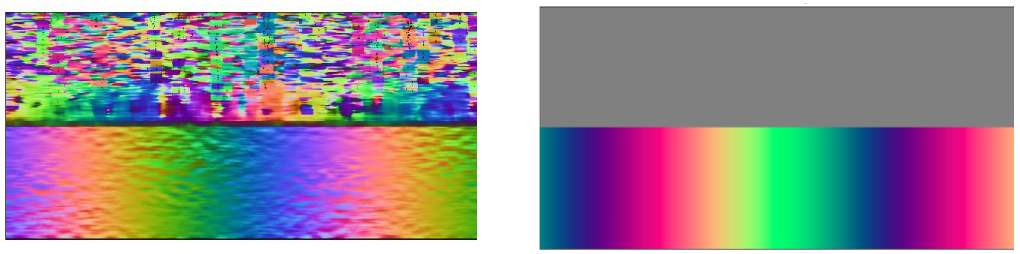


Figure: A simulated sample with and without PMD and speckle noise for a rectangular slab.

- **Goal**: To validate the polarization mode dispersion compensation method, we have developed a simulation environment to validate key parts of the method. We can simulate various geometries and input PMD and speckle noise. The simulation allows us to generate clean and noisy PMD images as shown in the Figure. These clean and noisy image pairs can be used to train a deep learning denoising network in a supervised manner on simulated data.

Contact: dr. Patrick González
(p.j.gonzalez@vu.nl)

Deep learning based segmentation and fine-tuning on OCT lung images

- **Description:** Optical Coherence Tomography (OCT) is a powerful imaging technique for imaging biological structures with high resolution. We use this technique, using self-designed, imaging catheters to study the airway smooth muscle of patients with lung diseases. However, due the design of the imaging catheter, the images contain artifacts hampering the clinical interpretation of these images.
- **Goal:** The goal of this project is to train, validate and implement a deep learning based segmentation network that can accurately identify image artifacts caused by the OCT imaging catheter design, i.e. by the catheter sheath and the motor wires. We will investigate both 2D and 3D networks based on the Unet architecture.

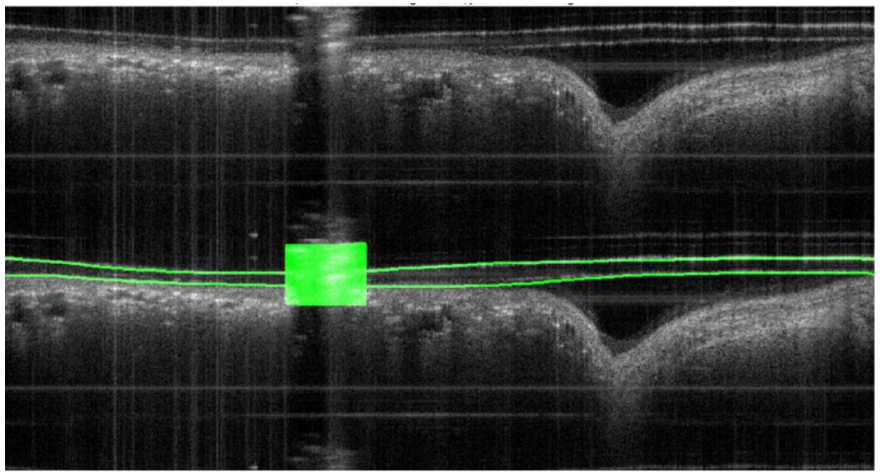


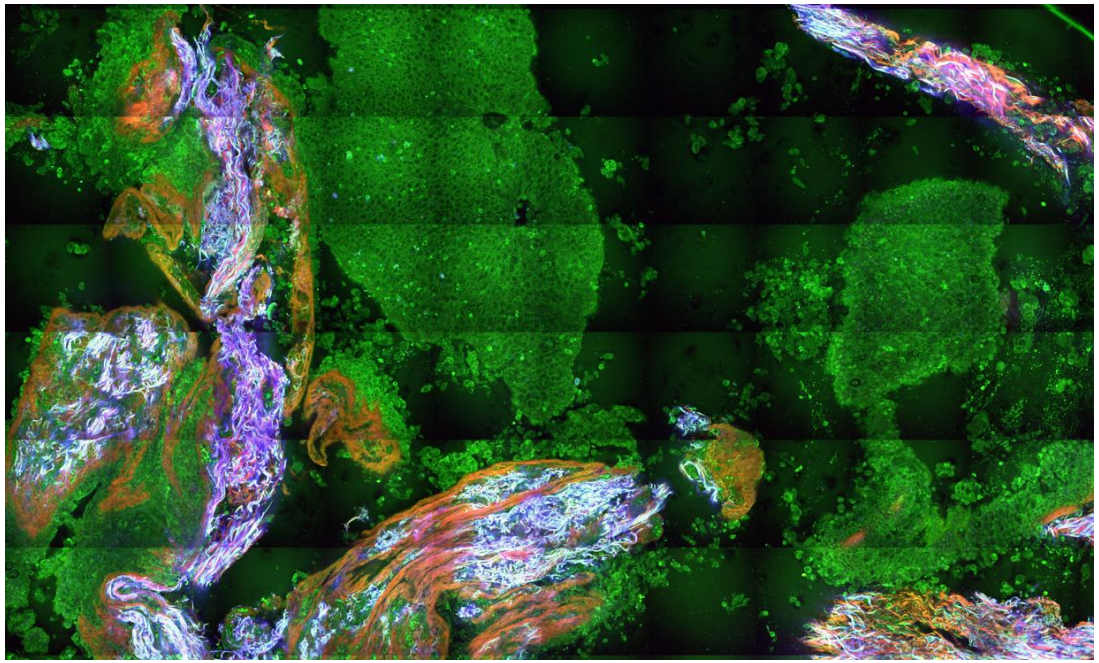
Figure: OCT-image, taken inside the lung area of a patient (upper half of picture), before the segmentation on this image has been done (lower half of picture, in green). The segmentation entails the Pebax-catheter (the two wire-like horizontal lines) as well as the distance between the vertical wires, marked as a block-like structure.

Contact: dr. Patrick González (p.j.gonzalez@vu.nl)

Image Analysis of Asthmatic Lung Biopsies

Description:

Asthma is associated with changes in the extracellular matrix (ECM) (mainly collagen & elastin), an increase in airway smooth muscle (ASM) and sometimes high levels of eosinophils. Higher harmonic generation (HHG) microscopy visualizes collagen, elastin, cells and other structures without any processing or staining.



Goal:

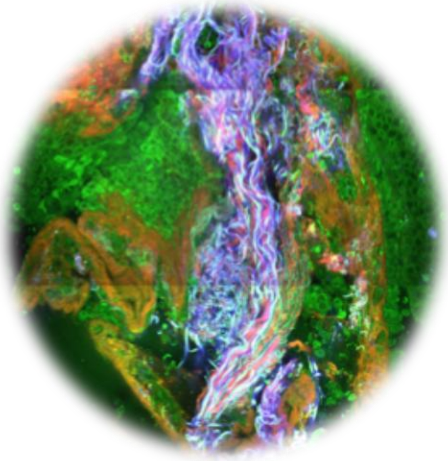
Quantification of the ASM, ECM and inflammatory components in asthma biopsies. Collagen and elastin are easy to identify by their different colour. However, the ASM and inflammatory cells need a different approach. (Annotation and AI-training)

Contact:
Prof. Dr. Marloes Groot: m.l.groot@vu.nl
Sylvia Spies: s.spies@vu.nl

Higher Harmonic Generation Endoscopy

Description:

Higher harmonic generation (HHG) microscopy visualizes collagen, elastin, cells and other structures without any processing or staining. Currently we are imaging only ex-vivo material. However, it is possible to apply this technique in-vivo using a flexible endoscope.



Goal:

There are multiple things that need to be improved, so there is a lot to choose from. Software improvements, better graphical user interface, adding extra detectors, improve acquisition timing

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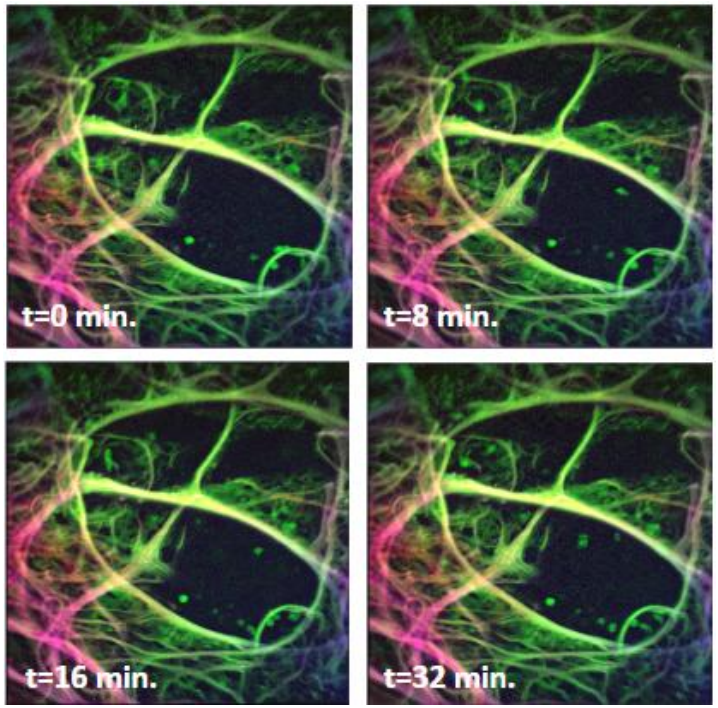
Prof. Dr. Marloes Groot: m.l.groot@vu.nl
Sylvia Spies: s.spies@vu.nl
Naomi Duits: n.e.duits@vu.nl

Image Analysis of timelapse videos of lung slices

Description:

Using higher harmonic generation, we are able to visualise different structures in the lung including immune cells, collagen and elastin. By generating timelapse videos, we can visualise dynamic processes in the alveolar structure of the lung, including movement of the immune cells. We want to analyse the dynamics of these immune cells.



Goal:

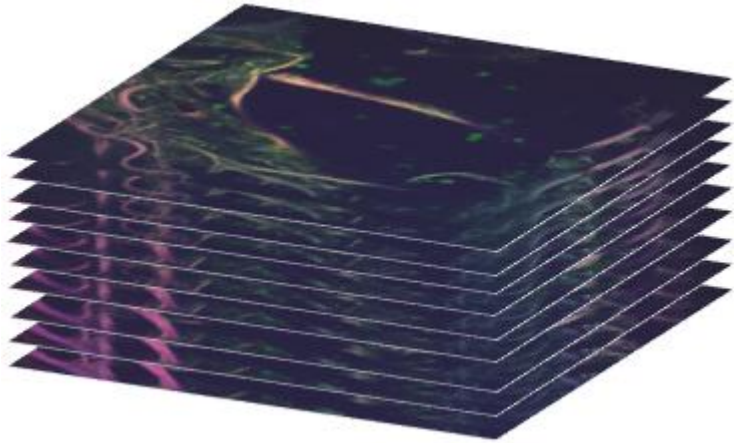
The goal of the project is to generate a workflow that applies simple image processing strategies to isolate and track the motion of the immune cells.

Contact:
Prof. Dr. Marloes Groot: m.l.groot@vu.nl
Naomi Duits: n.e.duits@vu.nl

Visualisation strategies for 3D HHG images (z-stacks)

Description:

Using higher harmonic generation, we are able to visualise different structures in tissue including cells, collagen and elastin. Through the generation of z-stack images, we can image 3D volumes of the scanned tissue. Currently, we visualise these volumes by scrolling trough the images one-by-one or by generating z-projections. A different approach might be more suitable for certain research questions.



Goal:

Investigate the added benefit and limits of different visualisation strategies for 3D HHG images.

Contact:
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Naomi Duits: n.e.duits@vu.nl